Standard form, and fractions as decimals

The national plan's "practical and interdisciplinary" lessons, carrying two Cambridge Stage 9 topics the programme does not otherwise reach.

LESSON 34–35 | 2 × 40 MIN | ALGEBRA 8, CHAPTER I PRACTICAL | STAGE 9 · 1.2, 8.1 [CAMBRIDGE INSERT]

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Write large and small numbers in standard form $a \times 10^n$, $1 \le a < 10$.
- Multiply and divide numbers written in standard form.
- Convert a fraction to a terminating or recurring decimal, and back.
- Tell a rational number from an irrational one by its decimal.

TEXTBOOK

National	pp. 62–67
Cambridge	Learner's Book pp. 14–16, 162–166
Workbook	Workbook 1.2, 8.1

01 Explanation

Standard form

DEFINITION

A number is in **standard form** when it is written as $a \times 10^n$ with $1 \le a < 10$ and n a whole number.

- $4\,500\,000 = 4.5 \times 10^6$ — the point moves 6 places left, so n is positive.
- $0.00032 = 3.2 \times 10^{-4}$ — the point moves 4 places right, so n is negative.

To calculate, handle the numbers and the powers of ten separately:

$$(3 \times 10^5) \cdot (2 \times 10^3) = 6 \times 10^8$$

$$(6 \times 10^5) : (3 \times 10^2) = 2 \times 10^3$$

TIDY THE ANSWER

40×10^4 is not in standard form. Rewrite it as 4×10^5.

Fractions as decimals

Divide the numerator by the denominator. Exactly one of two things happens:

- the division **stops** — a **terminating** decimal, e.g. $\frac{3}{8} = 0.375$;
- a block of digits **repeats for ever** — a **recurring** decimal, e.g. $\frac{1}{3} = 0.333\ldots = 0.(3)$ and $\frac{5}{11} = 0.454545\ldots = 0.(45)$.

WHICH ONE?

Cancel the fraction first. If the denominator's only prime factors are 2 and 5, the decimal terminates; otherwise it recurs. $\frac{7}{40}$ terminates ($40 = 2^3 \cdot 5$); $\frac{7}{30}$ recurs ($30 = 2 \cdot 3 \cdot 5$).

Every fraction gives a terminating or recurring decimal, and every terminating or recurring decimal comes from a fraction. A decimal that never stops *and* never repeats belongs to an **irrational** number — $\sqrt{2} = 1.41421356\ldots$, $\pi = 3.14159\ldots$.

Turning a recurring decimal back into a fraction

Multiply by a power of ten so that one whole period shifts past the point, then subtract.

$$x = 0.(7) \Longrightarrow 10x = 7.(7) \Longrightarrow 9x = 7 \Longrightarrow x = \frac{7}{9}$$

For a two-digit period multiply by 100: $x = 0.(45) \Longrightarrow 100x - x = 45 \Longrightarrow x = \frac{45}{99} = \frac{5}{11}$.

02 Worked examples

EXAMPLE 1 **Write 0.00047 and 92 000 in standard form.**

1. $0.00047 \to 4.7$

   The point moves 4 places right.

2. $= 4.7 \times 10^{-4}$

   Moving right gives a negative index.

3. $92\ 000 \to 9.2$

   The point moves 4 places left.

4. $= 9.2 \times 10^{4}$

ANSWER

4.7×10^{-4} and 9.2×10^{4}

EXAMPLE 2 ***Work out $(8 \times 10^{6}) : (4 \times 10^{2})$***

1. $8 : 4 = 2$

   The numbers.

2. $10^{6} : 10^{2} = 10^{4}$

   Subtract the indices.

3. $= 2 \times 10^{4}$

   Already in standard form.

ANSWER

$2 \times 10^{4} = 20\ 000$

EXAMPLE 3 ***Write 0.(6) as a fraction***

1. $x = 0.666\ldots$

   Name the number.

2. $10x = 6.666\ldots$

   One period is one digit, so multiply by 10.

3. $10x - x = 6$

   The tails cancel exactly.

4. $9x = 6,\ x = \dfrac{6}{9} = \dfrac{2}{3}$

   Cancel.

ANSWER

$$\frac{2}{3}$$

03 Interactive model

Ask for an estimate in standard form before any calculation — it catches most place-value slips.

Standard form and recurring decimals

0.00058

1. Move the point until one non-zero digit is in front:

5.8 .

That is the value of a.

2. The point moved

4 places right .

Right $\Longrightarrow$ negative index.

3. 5.8×10^{-4}

Check: 5.8 ÷ 10 000 = 0.00058 ✓

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Standard form	Standart ko‘rinish	Стандартный вид
Power of ten	O‘nning darajasi	Степень десяти
Significant figures	Ahamiyatli raqamlar	Значащие цифры
Terminating decimal	Chekli o‘nli kasr	Конечная десятичная дробь
Recurring decimal	Davriy o‘nli kasr	Периодическая десятичная дробь
Period	Davr	Период
Rational number	Ratsional son	Рациональное число
Irrational number	Irratsional son	Иррациональное число
Convert	Aylantirish	Преобразовать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

0.00062 in standard form is:

- 6.2×10^{-4}
- 6.2×10^{4}
- 62×10^{-5}
- 0.62×10^{-3}

$(2 \times 10^{3}) \cdot (5 \times 10^{4})$ equals:

- 10×10^{7}
- 1×10^{8}
- 7×10^{7}
- 10^{12}

Which fraction gives a recurring decimal?

- $\frac{3}{8}$
- $\frac{7}{20}$
- $\frac{5}{6}$
- $\frac{9}{25}$

$0.(3)$ as a fraction is:

- $\frac{3}{10}$
- $\frac{1}{3}$
- $\frac{3}{99}$
- $\frac{1}{30}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. *Write 5 000 in standard form*

   ANSWER 5×10^3

2. *Write 0.007 in standard form*

   ANSWER 7×10^{-3}

3. *Write 3.2×10^4 as an ordinary number*

   ANSWER 32 000

4. *Write 6×10^{-2} as an ordinary number*

   ANSWER 0.06

5. *Write $\frac{1}{4}$ as a decimal*

   ANSWER 0.25

6. *Write $\frac{1}{3}$ as a decimal*

   ANSWER 0.(3)

7. *Write 0.5 as a fraction in lowest terms*

   ANSWER $\frac{1}{2}$

Medium · the standard question

7 PROBLEMS

1. *Write 0.00047 in standard form*

   ANSWER 4.7×10^{-4}

2. *Write 92 000 in standard form*

   ANSWER 9.2×10^{4}

3. *Work out* $(3 \times 10^{5}) \cdot (2 \times 10^{3})$

   ANSWER 6×10^{8}

4. *Work out* $(8 \times 10^{6}) : (4 \times 10^{2})$

   ANSWER 2×10^{4}

5. *Does* $\frac{7}{40}$ *terminate? Give the decimal.*

   ANSWER Yes — 0.175

6. *Write* $\frac{5}{11}$ *as a decimal*

   ANSWER $0.(45)$

7. *Write* $0.(6)$ *as a fraction*

   ANSWER $\frac{2}{3}$

Hard · stretch and reason

7 PROBLEMS

1. *Work out* $(4 \times 10^5) \cdot (5 \times 10^4)$ *in standard form*

   ANSWER 2×10^{10}

2. *Work out* $(1.2 \times 10^7) : (4 \times 10^3)$

   ANSWER 3×10^3

3. *Write 0.(45) as a fraction in lowest terms*

   ANSWER $\frac{5}{11}$

4. *Write 0.1(6) as a fraction*

   ANSWER $\frac{1}{6}$ — multiply by 10 and by 100, then subtract.

5. *Without dividing, say whether* $\frac{9}{48}$ *terminates*

   ANSWER Cancel to $\frac{3}{16}$; $16 = 2^4$, so yes.

6. *The mass of a grain of sand is about* 6.6×10^{-5} *kg. Find the mass of* 2×10^6 *grains.*

   ANSWER $1.32 \times 10^2 = 132$ kg

7. *Is 0.101001000100001… rational? Explain.*

   ANSWER No — it never stops and never repeats, so it is irrational.

07 Homework

Homework — 6 problems

Algebra 8, pp. 62–67 (practical problems) and Cambridge Workbook 1.2.

1. *Write 0.000091 and 340 000 in standard form*
2. *Work out* $(5 \times 10^4) \cdot (6 \times 10^3)$ *in standard form*
3. *Work out* $(9 \times 10^8) : (3 \times 10^5)$

4. *Write* $\frac{7}{8}$ *and* $\frac{4}{9}$ *as decimals*

5. *Write 0.(8) as a fraction*

6. *Without dividing, decide whether* $\frac{11}{50}$ *terminates*